

Exam Date & Time: 04-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Reinforcement Learning [NPT-4006]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Develop tic-tac-toe game using generalized approach by using a 9-element array representing the board. While developing the game list out and draw all the board positions which could result the next moves until win or draw appears on the board. (5)
 - A)
 - B) Illustrate the advantages of generalized approach in developing tic-tac-toe game. (5)
- 2) Find the mean value for the continuous random variable, $f(x) = \sqrt{x}, 1 \leq x \leq 9$ (5)
 - A)
 - B) The PDF of a continuous random variable X is $f(x) = \begin{cases} \theta e^{-\theta x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$ With $\theta > 0$. Find the cumulative distribution function F(u) (3)
 - c) Prove if $A \subset B$ then $P(A) \leq P(B)$ (2)
- 3) You are a data analyst at a fitness app that shows personalized exercise plans to its users. The app has five different workout plans it can recommend as the Plan of the Day: Plan A, Plan B, Plan C, Plan D, and Plan E. When users open the app, they see one of these workout plans, and if they start the workout, it counts as a "success." The objective is to maximize the start rate of workout plans by finding which plan resonates the most with users. However, the performance of each plan is unknown, so you need a strategy to balance exploring different plans to gather data and exploiting the best-performing plan to boost user engagement. Make use of epsilon-greedy algorithm to decide which workout plan to show to users (5)
 - A)
 - B) Use the above scenario in question 3(a) and write the steps involved in implementation of epsilon-greedy approach. (5)
- 4) Explain in detail how Dynamic Programming is used in Reinforcement Learning and its significance in solving decision-making problems. (5)
 - A)
 - B) Discuss how an optimal policy will be derived and evaluated in reinforcement learning. (5)

Elaborate with an example.

- 5) Imagine you are developing a game-playing agent for a board game similar to chess. The agent must learn optimal strategies to win by exploring different moves and their outcomes. To evaluate its performance, you decide to use the Monte Carlo method for policy evaluation. As the agent plays numerous games, it will record the sequence of states, actions, and rewards from each game. After several iterations of playing against itself or other opponents, you want to analyze how the Monte Carlo method can help improve the agent's policy based on the outcomes of these games. In the context of the board game agent using the Monte Carlo method, Discuss the steps involved in implementing policy evaluation. How does the Monte Carlo method help the agent learn from its experiences, and what are the potential challenges associated with using this approach in dynamic environments? (5)
- A)
- B) Explain the significance of Actor-Critic methods in reinforcement learning and discuss their mechanisms and applications. (5)

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